

What's New in LEF/DEF 5.8

Product Version 5.8

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About This Manual

This document provides information on new and changed features for version 5.8 of the Cadence® Library Exchange Format (LEF) and Design Exchange Format (DEF) integrated circuit description languages.

How This Document Is Organized

This *What's New* document is organized into the following chapters:

- **New Features**

This chapter describes features that were added or changed since version 5.7 of LEF and DEF.

New features are those that introduce new functionality into LEF and DEF. Any enhancements made to existing statements to support a new feature are also described in this chapter.

Related Documents

The following documents provide detailed information about LEF and DEF, and the LEF and DEF application programming interfaces.

- [LEF/DEF Language Reference](#)

New Features

- [Multi-Mask Patterning Technology](#)
- [New LEF Features](#)
- [New DEF Features](#)

Multi-Mask Patterning Technology

The multi-mask patterning technology allows you to define multi-mask patterns for layers. You can define multi-mask statements that split a single layer into two or three masks. The syntax enhancements made to LEF/DEF v5.8 support more than three masks, though most applications support two or three masks only.

The subsequent sections describe the new syntax changes that have been made to incorporate this new feature.

New LEF Features

The following sections describe the enhancements that have been made to support multi-mask patterning for LEF.

- [LEF Layers Support Masks](#)
- [Fixed Masks for Layers and Macros](#)
- [LEF Via Adds Masks for Fixed Vias](#)
- [Macro Layer Geometries Support Masks](#)

LEF Layers Support Masks

The `MASK` statement was added to specify how many masks for double- or triple-patterning will be used for a specified layer. You can use this statement for layers - cut, routing, implant, overlap, and/or masterslice.

Includes the following new syntax (in bold):

```
LAYER layerName  
  
TYPE ROUTING ;                #or TYPE CUT, IMPLANT, MASTERSLICE, OVERLAP  
  
[MASK numMasks ;]
```

Fixed Masks for Layers and Macros

Some technologies do not allow mask shifting for cells using multi-mask patterning, e.g., pin and routing shapes are all pre-colored and must not be shifted to other masks. The `FIXEDMASK` statement was added to indicate that mask shifting is not allowed. All the LEF macro pin mask assignments are kept fixed and cannot be shifted to a different mask.

Includes the following new syntax (in bold):

For layers:

```
FIXEDMASK ;  
  
LAYER <statement>
```

For macros:

```
MACRO macroName  
  
[CLASS className [subclassName] ;]
```

```
[FIXEDMASK ;]
```

```
...
```

LEF Via Adds Masks for Fixed Vias

For fixed vias - rectangle or polygon - the masks in the via-master for cut layer are required for multi-cut vias. The `VIARULE` statement was enhanced to include masks. The `MASK` statement indicates which mask for double- or triple-patterning lithography is to be applied to the shapes defined using the `RECT` or `POLYGON` statements of the via master.

Includes the following new syntax (in bold):

```
VIA viaName [DEFAULT]

{ VIARULE viaRuleName ;

    CUTSIZE xSize ySize ;

    LAYERS botMetalLayer cutLayer topMetalLayer ;

    CUTSPACING xCutSpacing yCutSpacing ;

    ENCLOSURE xBotEnc yBotEnc xTopEnc yTopEnc ;

    [ROWCOL numCutRows numCutCols ;]

    [ORIGIN xOffset yOffset ;]

    [OFFSET xBotOffset yBotOffset xTopOffset yTopOffset ;]

    [PATTERN cutPattern ;]}

| [RESISTANCE resistValue ;]

{LAYER layerName ;

    { RECT [MASK maskNum] pt pt ;

      | POLYGON [MASK maskNum] pt pt pt ...; } ...

    }...

}

[PROPERTY propName propVal ;] ...

END viaName
```

Macro Layer Geometries Support Masks

Macro layer geometries support multi-mask patterning. Three different masks can be added to the same layer geometries. The `MASK` statement was added to indicate which mask from double- or triple-patterning is to be used for path, rectangle, polygon, or via geometries.

Includes the following new syntax (in bold):

```
{ LAYER layerName

    [EXCEPTPGNET]

    [SPACING minSpacing | DESIGNRULEWIDTH value] ;

    [WIDTH width ;]

    { PATH [MASK maskNum] pt ... ;

    | PATH [MASK maskNum] ITERATE pt ... stepPattern ;

    | RECT [MASK maskNum] pt pt ;

    | RECT [MASK maskNum] ITERATE pt pt stepPattern ;

    | POLYGON [MASK maskNum] pt pt pt pt ... ;

    | POLYGON [MASK maskNum] ITERATE pt pt pt pt ... stepPattern ;

    } ...

    | VIA [MASK viaMaskNum] pt viaName ;

    | VIA ITERATE [MASK viaMaskNum] pt viaName stepPattern ;

    } ...
```

For syntax information, see the [LEF Syntax](#) chapter in the *LEF/DEF Language Reference*.

New DEF Features

The following sections describe the enhancements that have been made to support multi-mask patterning for DEF.

- [Routing Blockages Add Mask Information](#)
- [Component Mask Shift Multi-Patterning Layers](#)
- [COMPONENTS Adds Mask Shift Information](#)
- [FILLS Adds Masks](#)
- [NETS Routing Points Enhancements](#)
- [Special Nets Special Wiring Enhancements](#)
- [DEF VIAS Adds Mask](#)
- [TRACKS Adds Mask](#)
- [PINS Adds Mask](#)

Routing Blockages Add Mask Information

The `BLOCKAGES` section was enhanced to include masks. The `MASK` statement indicates which mask for double- or triple-patterning lithography is to be used for the specified shapes.

Includes the following new syntax (in bold):

```
[BLOCKAGES numBlockages ;  
  
  [- LAYER layerName  
  
    [+ COMPONENT compName | + SLOTS | + FILLS | + PUSHDOWN  
      | + EXCEPTPGNET]  
  
    [+ SPACING minSpacing | + DESIGNRULEWIDTH effectiveWidth]  
  
    [+ MASK maskNum]  
  
      {RECT pt pt | POLYGON pt pt pt ...} ...  
  
  ;] ...  
  
END BLOCKAGES]
```

Component Mask Shift Multi-Patterning Layers

The `COMPONENTMASKSHIFT` statement defines which layers of a component are allowed to be shifted from the original mask colors in the LEF. This can be useful to shift all the layers of a specific component in order to align the masks with other component or router mask settings to increase routing density. This definition allows a specific component to compactly describe the mask shifting for that component.

Includes the following new syntax:

```
[COMPONENTMASKSHIFT layer1 [layer2 ...] ;]
```

COMPONENTS Adds Mask Shift Information

Some applications require the ability to "shift" the mask of pins and obstructions on a certain layer for a specific component to better align the pin masks with the routing masks.

The `MASKSHIFT` statement was added to allow shifting of the cell-master masks used in double- or triple-patterning for specific layers of an instance of the cell-master.

Includes the following new syntax (in bold):

```
COMPONENTS numComps ;

  [- compName modelName

    [+ EEQMASTER macroName]

    [+ SOURCE {NETLIST | DIST | USER | TIMING}]

    [+ {FIXED pt orient | COVER pt orient | PLACED pt orient
      | UNPLACED} ]

    [+ MASKSHIFT shiftLayerMasks]

    [+ HALO [SOFT] left bottom right top]

    [+ ROUTEHALO haloDist minLayer maxLayer]

    [+ WEIGHT weight]

    [+ REGION regionName]

    [+ PROPERTY {propName propVal} ...]...

  ;] ...

END COMPONENTS
```

FILLS Adds Masks

In order to support multi-patterning technologies up to three different masks for the same layer can be added.

The `MASK` statement was added to specify which mask for double- or triple-patterning lithography is to be applied to the specified rectangles, polygons, or via shapes on each layer.

Includes the following new syntax (in bold):

```
[FILLS numFills ;  
  
  [- LAYER layerName [+ MASK maskNum] [+ OPC]  
  
    {RECT pt pt  
  
      | POLYGON pt pt pt ...} ... ;] ...  
  
  [- VIA viaName [+ MASK viaMaskNum] [+ OPC] pt ... ;] ...  
  
END FILLS]
```

NETS Routing Points Enhancements

NETS Adds Rectangle and/or Virtual Connection

The `NETS` section was enhanced to provide the ability to create a rectangle or add a virtual connection to maintain the symbolic routing graph.

The `RECT` statement allows creating a rectangle from the previous routing points. The `VIRTUAL` statement indicates that there is a virtual connection between the previous point and the new routing point.

Includes the following new syntax (in bold):

```
{ ( x y [extValue] )  
  
  { ( x y [extValue] )  
  
    | viaName [orient]  
  
    | RECT ( deltax1 deltay1 deltax2 deltay2 )  
  
    | VIRTUAL ( x y ) } } ...
```

NETS Adds Masks

The `NETS` section routing points was enhanced to add masks - up to three different masks for the same layer can be added.

Includes the following new syntax (in bold):

```
{ ( x y [extValue] )  
  [MASK maskNum] ( x y [extValue] )  
  | [MASK viaMaskNum] viaName [orient]]  
  | [MASK maskNum] RECT ( deltax1 deltax2 deltay1 deltay2 )  
  | VIRTUAL ( x y ) } } ...
```

Special Nets Special Wiring Enhancements

To support multi-mask patterning up to three different masks for the same layers can be added to special nets.

A new `VIA` statement was added to make it more efficient to specify several power-vias.

Additionally, the route status - cover, fixed, routed, shield, and shape - can be applied to polygons, rectangles, and vias.

Includes the following new syntax (in bold):

```
[ [+ COVER | + FIXED | + ROUTED | + SHIELD shieldNetName]  
  [+ SHAPE shapeType] [+ MASK maskNum]  
  + POLYGON layerName pt pt pt...  
  | + RECT layerName pt pt  
  | + VIA viaName [orient] pt ...  
| {+ COVER | + FIXED | + ROUTED | + SHIELD shieldNetName}  
  
  layerName routeWidth  
  
    [+ SHAPE shapeType]  
  
    [+ STYLE styleNum]  
  
    routingPoints  
  
[NEW layerName routeWidth
```

```
[+ SHAPE shapeType]  
  
[+ STYLE styleNum]  
  
routingPoints  
  
] ...  
  
] ...
```

New syntax for routing points:

```
(x y [extValue] )  
  
{ [MASK maskNum] (x y [extValue] )  
  
| [MASK viaMaskNum] viaName [orient]  
  
[DO numX BY numY STEP stepX stepY]  
  
}...
```

DEF VIAS Adds Mask

For fixed vias, rectangle or polygon, the masks in the via-master for cut-layer shapes are required for multi-cut vias. The `MASK` statement indicates which mask for double- or triple-patterning lithography is to be applied to the defined shapes.

Includes the following new syntax (in bold):

```
[VIAS numVias ;  
  
[- viaName  
  
{ + VIARULE viaRuleName  
  
+ CUTSIZE xSize ySize  
  
+ LAYERS botmetalLayer cutLayer topMetalLayer  
  
+ CUTSPACING xCutSpacing yCutSpacing  
  
+ ENCLOSURE xBotEnc yBotEnc xTopEnc yTopEnc  
  
[+ ROWCOL numCutRows NumCutCols]  
  
[+ ORIGIN xOffset yOffset]  
  
[+ OFFSET xBotOffset yBotOffset xTopOffset yTopOffset]  
  
[+ PATTERN cutPattern]
```

```
    | [ + RECT layerName [+ MASK maskNum] pt pt

    | + POLYGON layerName [+ MASK maskNum] pt pt pt ...

    }

;] ...

END VIAS]
```

TRACKS Adds Mask

The routing tracks can now have mask assignments to aid alignment of routing masks inside a block with the level above. The `MASK` statement indicates which mask for double- or triple-patterning lithography to use for the first routing track.

Includes the following new syntax (in bold):

```
[TRACKS

    [{X | Y} start DO numtracks STEP space

        [MASK maskNum [SAMEMASK] ]

        [LAYER layerName ...]

    ;] ...]
```

PINS Adds Mask

The `PINS` syntax has been modified to include mask attributes. The `MASK` statement indicates which mask from double- or triple-patterning to use for pins or via shapes on each layer.

Includes the following new syntax (in bold):

```
[PINS numPins ;

    [ [- pinName + NET netName]

    ...

    [[+ PORT]

        [+ LAYER layerName

            [MASK maskNum]

            [SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]

            pt pt
```

```
|+ POLYGON layerName

    [MASK maskNum]

    SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]

    pt pt pt ...

|+ VIA viaName

    [MASK viaMaskNum]

    pt

] ...

[+ COVER pt orient | FIXED pt orient | PLACED pt orient]

]...

; ] ...

END PINS]
```

For syntax information, see the [DEF Syntax](#) chapter in the *LEF/DEF Language Reference*.